

# ANAM MYOLWA

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## Professional Summary

Versatile Data Engineer and Software Developer, and MSc Computer Science candidate specializing in Software Engineering. Demonstrated expertise in building secure, scalable enterprise applications and clinical data platforms, with current experience developing ETL pipelines, REDCap dashboards, and OpenMRS solutions for health information systems. Proficient in the complete SDLC from requirements analysis through deployment, with hands-on experience in DevOps practices, containerization, and CI/CD automation. Strong foundation in data structures, algorithms, machine learning, and cryptography. Proven ability to translate complex technical concepts into actionable solutions while maintaining security compliance and system reliability. Effective communicator and collaborative team player with leadership experience in mentoring students.

## Core Competencies

- Full-Stack Web Development
- API Design & Microservices Architecture
- Cloud Computing & Infrastructure (Azure)
- Database Design & Optimization
- DevOps & CI/CD Pipeline Management
- Container Orchestration (Docker, Kubernetes)
- Machine Learning & Data Analytics
- Agile/Scrum Methodologies
- Technical Documentation & Communication
- Team Leadership & Mentoring

## Professional Experience

### Data Engineer (Fixed-Term Contract) | BLOODSA, Groote Schuur Hospital

*Observatory, Cape Town | July 2026 – December 2026*

- Lead the design and development of clinician-facing REDCap dashboards covering patient volumes per clinician and clinic day, identification of missing or incomplete data, task tracking, and patient-level action lists
- Design, build, and maintain scalable pipelines to extract, transform, and integrate data from multiple health information systems into REDCap and OpenMRS
- Support the implementation and customisation of OpenMRS as a clinical data platform, including structured clinical forms, workflow optimisation for outpatient haematology clinics, and configuration of patient tracking, appointments, and longitudinal records
- Architect robust data integration solutions, including API development and system interoperability between REDCap, OpenMRS, and external systems such as the NHLS
- Lead the migration of legacy registry data into HOPeR and OpenMRS through data mapping and harmonisation, validation and reconciliation, and quality-assurance testing
- Develop automated data capture, workflows, and notification systems (laboratory result integration, email alerts, task triggers) to reduce manual data entry burden
- Contribute to system architecture planning, produce high-quality technical documentation, and provide training and mentorship within BLOODSA while collaborating with clinicians, researchers, and operational teams

### Software Developer Intern | Department of Sport, Arts and Culture

*Pretoria, Gauteng | May 2024 – May 2026*

- Developed and maintained web applications using Django, HTML, CSS, and JavaScript to support programme management and event coordination
- Designed database schemas in Microsoft SQL Server and PostgreSQL, optimizing queries for reporting and data retrieval
- Built backend features including user authentication, session management, and form validation to improve data accuracy
- Implemented security measures such as input sanitization, CSRF protection, and secure coding practices across production systems
- Identified and resolved bugs through systematic testing and debugging, improving overall system reliability
- Deployed applications on Linux servers, configuring Nginx, managing static files, and setting up production environments
- Created technical documentation to support knowledge transfer and onboarding for future team members
- Collaborated with stakeholders to gather requirements and deliver features within Agile sprint cycles

### Computer Science Tutor & Teaching Assistant | University of Fort Hare

*Alice, Eastern Cape | March 2023 – December 2023*

- Conducted hands-on lab sessions covering algorithm design, complexity analysis, software testing, debugging techniques, and version control systems, strengthening students' problem-solving abilities and practical coding skills through real-world projects.

- Designed supplementary learning materials, coding exercises, and assessment rubrics aligned with curriculum objectives, while providing constructive feedback on assignments and fostering peer collaboration through study groups and code review sessions.
- Delivered technical training workshops on Microsoft Office Suite, LaTeX document preparation, and academic research methodologies, enhancing digital literacy and preparing students for professional and academic success with 90% satisfaction ratings.

## Technical Projects

### #053 Schools Junior League Management System [www.053schoolsjuniorleague.online/](http://www.053schoolsjuniorleague.online/)

- Sole developer of a full-stack web application for an active junior football league, built with Django and deployed on PythonAnywhere. The system manages player registrations, live standings, fixture scheduling, and result tracking across six age divisions and eight participating schools.
- Features include a staff admin panel for verifying players, entering match results, posting announcements, and uploading documents, as well as a public-facing portal for parents and coaches to register players, follow standings, and download league resources.
- *Technologies: Python, Django, MySQL, HTML, CSS, JavaScript, PythonAnywhere*

### Real-Time Data Analytics Dashboard for Government Reporting

- Built an ETL pipeline using Python (pandas, Apache Airflow) to extract, transform, and load data from multiple government databases (SQL Server, Oracle) into a centralized data warehouse for business intelligence reporting.
- Created interactive Power BI dashboards with custom DAX measures for executive-level reporting on program performance, budget allocation, and citizen engagement metrics, serving 200+ government stakeholders.
- *Technologies: Python, Apache Airflow, SQL Server, Oracle Database, Power BI, DAX, REST APIs*

### Secure Student Management System with Role-Based Access Control

- Developed a comprehensive student information system using Java Spring Boot and Angular, implementing RBAC for administrators, faculty, and students with granular permission management and audit logging.
- Integrated LDAP authentication implemented AES-256 encryption for sensitive data at rest, and deployed on Tomcat with SSL/TLS certificates for secure data transmission, ensuring GDPR and POPIA compliance.
- *Technologies: Java, Spring Boot, Angular, MySQL, LDAP, Tomcat, JUnit, Hibernate ORM*

## Education

### Master of Science in Computer Science – Software Engineering

*University of Fort Hare, Alice, Eastern Cape | In Progress (Expected 2027)*

### Bachelor of Science (Honours) in Computer Science

*University of Fort Hare, Alice, Eastern Cape | December 2023*

### Bachelor of Science in Computer Science

*University of Fort Hare, Alice, Eastern Cape | December 2022*

## Technical Skills

**Languages (Expert):** Python, Java, JavaScript/TypeScript, SQL, C#, C/C++, Bash/Shell, PowerShell, R

**Web Frameworks:** Django, Flask, FastAPI, Spring Boot, React.js, Angular, Node.js, Express.js, Bootstrap, Tailwind CSS

**Databases:** PostgreSQL, MySQL, Microsoft SQL Server, Oracle Database, MongoDB, Redis, SQLite, Database Design & Normalization

**DevOps & Cloud:** Docker, Kubernetes, Jenkins, GitLab CI/CD, GitHub Actions, Azure (VMs, App Services, DevOps), Terraform, Ansible, Nginx, Apache

**Data Science & ML:** pandas, NumPy, scikit-learn, TensorFlow, Keras, PyTorch, Matplotlib, Seaborn, Jupyter, Apache Spark

**IDEs & Editors:** VS Code, IntelliJ IDEA, PyCharm, Eclipse, Visual Studio, Vim, Sublime Text, Jupyter Notebooks

**Operating Systems:** Linux (Ubuntu, CentOS, Kali), Windows Server, macOS, Active Directory, Group Policy, System Administration

## Certifications

- Google Business Intelligence, Google, 2025
- CompTIA Security+, Alison, 2025
- CompTIA A+, Alison, 2025
- Microsoft Azure Administrator, Alison, 2025
- Google IT Support, Google, 2025
- Data Science Essentials with Python, CISCO, 2026
- Advanced Network Security, LearnQuest, 2025
- Complete Linux Training, Packt, 2025
- ITIL, Alison, 2025
- Introduction to Cybersecurity, CISCO, 2023
- Networking Basics, CISCO, 2026